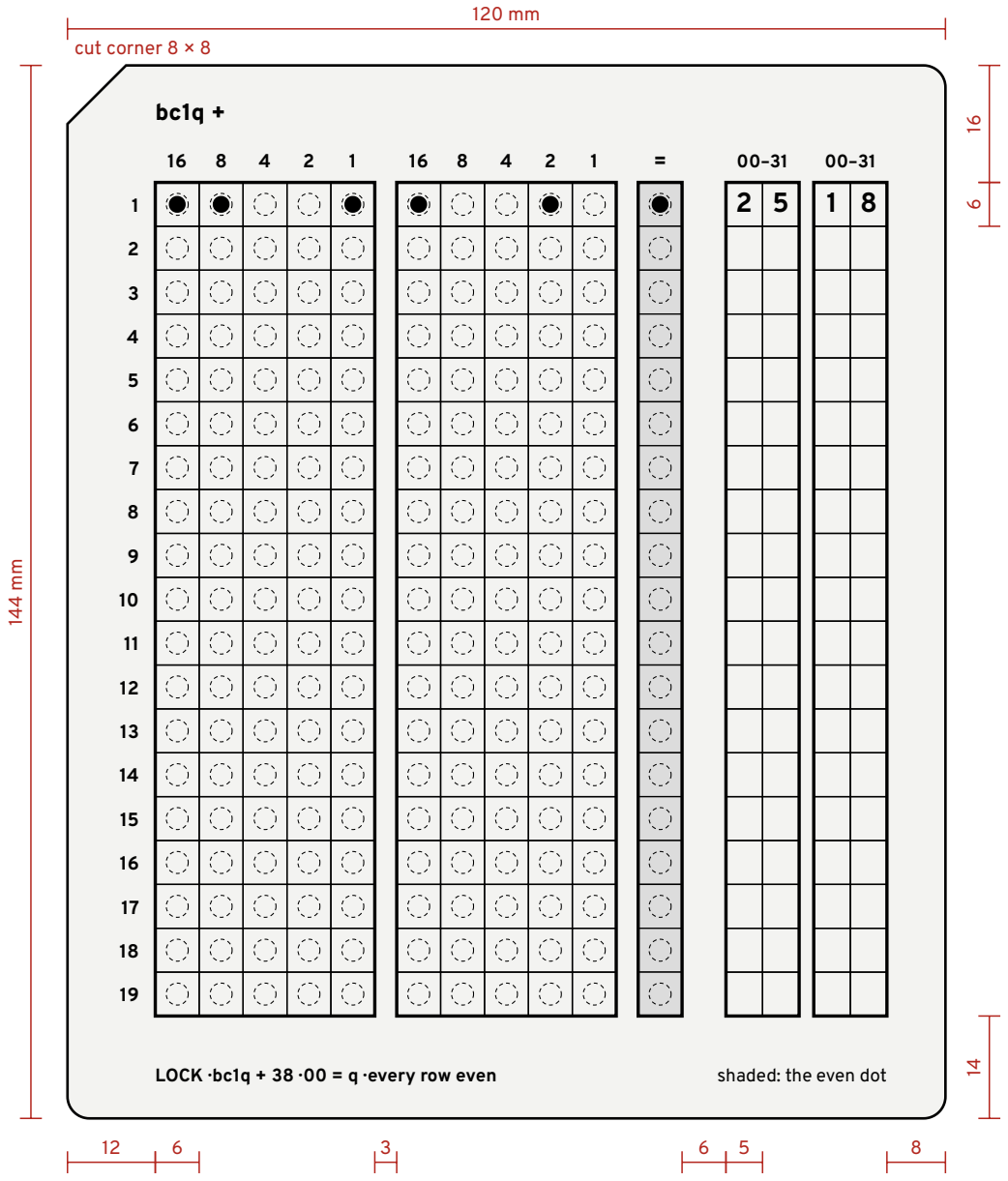
KEY lock plate ·design drawing, full size

The public half of the key: the one address it will ever receive to. LOCK shows it on the lights as nineteen rows, and each row is punched and stamped the same way as a row of the seed plate. This plate replaces the xpub plate.



What a row holds

The address is **bc1q** and 38 characters. Each character is a number from 00 to 31, which is five dots. A row holds two of them, plus one dot that makes the row's dot count even. Row 1 shows 25 and 18.

Made in advance

The grid, row numbers 1 to 19, the weights 16 8 4 2 1 over each group of five, the legend, and a cut corner at top **left**. The seed plate's is top right, so the two cannot be mixed up.

Made by hand

Dots: copied from LOCK's lights, light for light. **Digits:** add the weights of the dots in each group and stamp the two numbers. Number stamps only.

Three checks

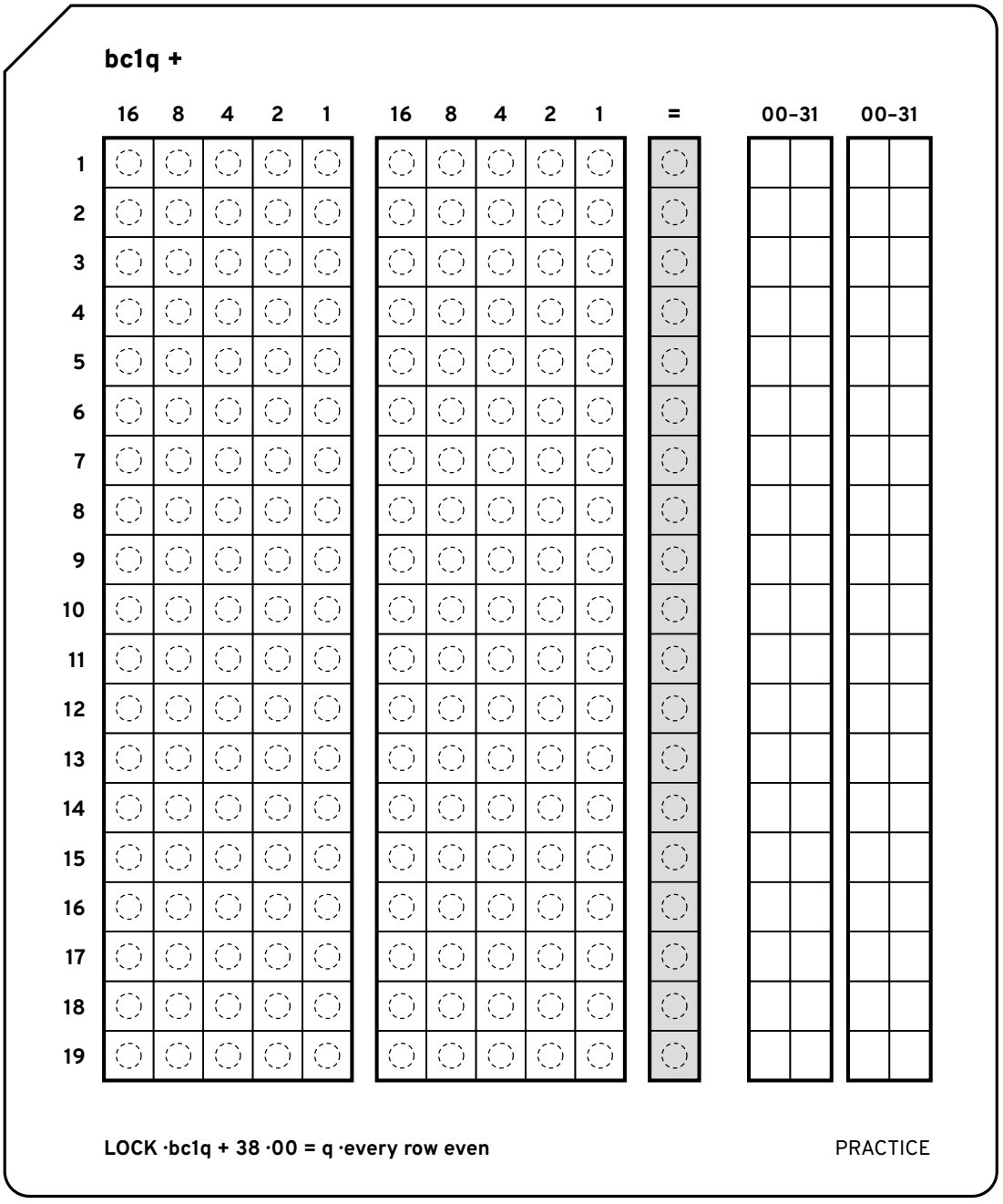
Every row has an even number of dots. Dots agree with digits. And the address's last six characters are its own checksum, so a wallet refuses a wrong one.

Still to decide

The plate is 120 × 144 mm, taller than the seed plate (120 × 100 mm). Whether the address also goes on the outside as a QR code, made from the written address by any public tool. Whether the address card is also engraved on the back of this plate, so the plate explains itself. Drawn for **3 mm number stamps**, an assumption. Print at 100%: the top edge measures 120 mm.

KEY lock plate ·paper, for the dry run

Copy each row from LOCK's lights: a filled dot for each light that is on. Then add the weights in each group of five and write the two numbers. Drawn a little larger than the steel plate, for pen work.



Before going on: every row has an even number of dots, and every number is between 00 and 31.

The address

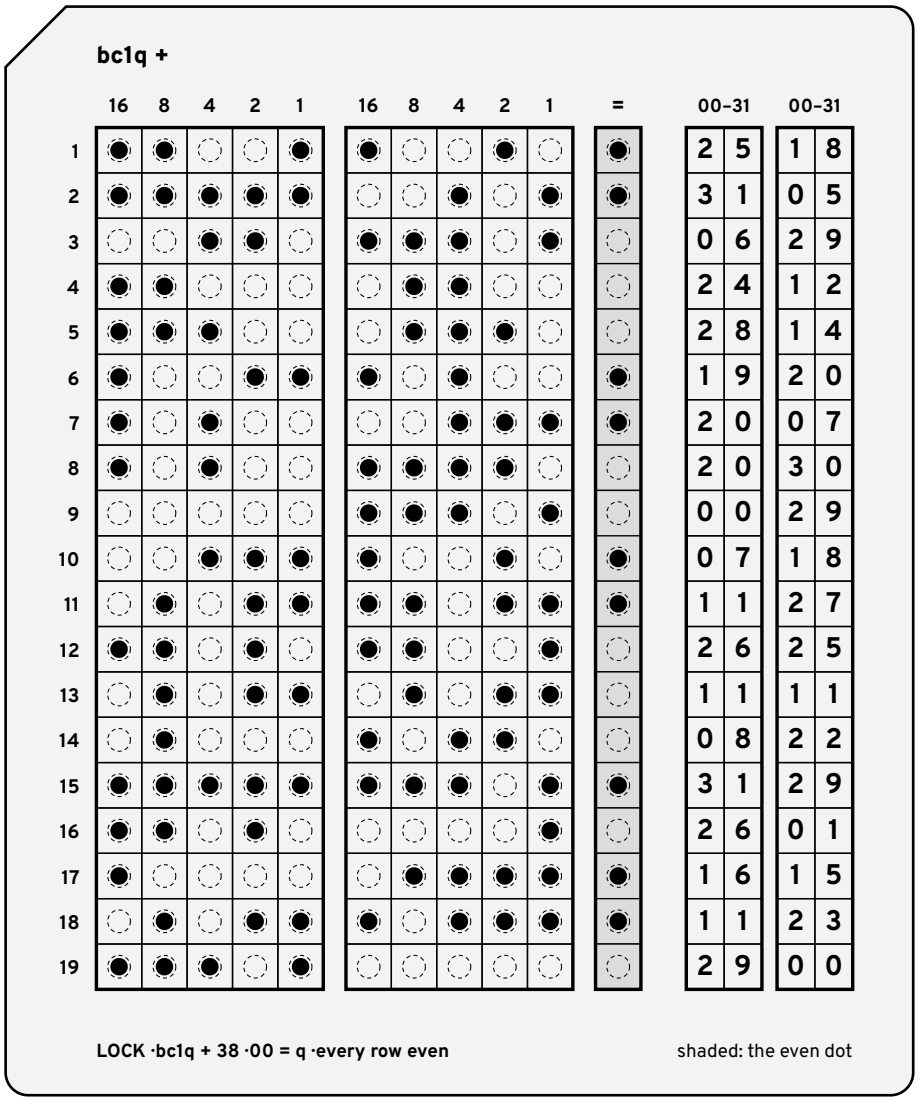
Write **bc1q**, then look up each number on the address card (a separate sheet), row by row, left number first.

bc1q

Type the address into a wallet as a watch-only address. If a character is wrong, the wallet will refuse it. Destroy this sheet when the dry run is finished.

KEY lock plate ·worked example

The lock for the published test key on page 3 of the seed plate design. The two worked examples are a pair: that key, this lock. It is a test pattern: never send money to it.



Read it

Row 1: the first group has dots under 16, 8 and 1, which is **25**. The second has dots under 16 and 2, which is **18**. On the address card, 25 is **e** and 18 is **j**.

That makes five dots, so the shaded cell has a sixth to make the row even.

The address

```
bc1q ejl9 xacv uwn5
5857 qa8j tm6e ttgk
la6p s0th aq
```

Scan it

A phone should read exactly the address above. This code was made from the written address by an ordinary program: nothing about it is secret.

To check the stand-in: run it with the eleven numbers of the seed plate example and the bits 1010010. Its nineteen rows must equal this plate.